

Raj Shekhar

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Education

Indian Institute of Technology Patna (IIT Patna)

2023 – 2027

B.Tech. in Mechanical Engineering

4th Year

Technical Skills

Programming: Python, C++, TypeScript, JavaScript, SQL, HTML, CSS

Web Development: React, Next.js, FastAPI, Flask, Streamlit, SQLAlchemy, REST APIs

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Cross-Validation, scikit-learn, XGBoost, LightGBM, SHAP, PyTorch, pandas, NumPy

Computer Vision & Medical Imaging: Image Processing, Pattern Recognition, OpenCV, Deep Learning, CNN, U-Net, Semantic / Instance Segmentation, Object Detection, YOLOv11, Multi-Object Tracking, Digital Pathology, High-resolution WSI Tiling, Quantification, ONNX

NLP & Retrieval: RAG, Embeddings, Semantic Search, Vector Search, FAISS, Cosine Similarity, Structured Outputs, Transformers, Generative AI, LLMs

Databases & Cloud: PostgreSQL, SQLite, Redis, pgvector, Vector Databases, Docker, Render

Tools & Engineering: Git, GitHub Actions, MLflow, JWT, WebSockets, pytest, PyMuPDF

Experience

Lattice Circularity Analyzer

Jun 2026 – Present

Research Intern | IIT Patna

[GitHub](#) | [Live](#)

- Predicted EDM hole circularity with Gradient Boosting (120 trees) plus a physics heuristic, using 16 SEM-labeled lab runs and 1,100 synthetic points; Gaussian Process (Matérn 5/2) LOOCV MAE ≈ 0.5 on a 1–5 scale.
- Engineered 15+ EDM and geometry features (current, pulse-on, duty, discharge energy, strut/node distances, overlap fractions) for a 900 μm tool that is $3.82 \times$ a 235.6 μm pore.
- Flagged PASS only when circularity $\geq 3.5/5$, ratio ≥ 0.70 , and geometry risk ≤ 0.55 — matching the lab finding that only 1 of 16 SEM trials kept an intact circular ring.

AetherShield VMS

2026

AI / Computer Vision | AI Fence

[GitHub](#) | [Live](#)

- Built a real-time video-management platform for object detection, tracking, identity classification, and zone-intrusion monitoring; deployed YOLOv11n + ByteTrack at 0.45 confidence / 200 ms, tracking 11 COCO classes across rectangle, circle, and polygon ROIs.
- Integrated Haar-cascade face matching with 64×64 histogram embeddings and cosine similarity threshold 0.72 for known / unknown / blacklist identity classification on person tracks.
- Enabled natural-language video search and video QA using structured detection metadata (class, track ID, zone, timestamp) instead of raw video frames.

SCHOLAR

May 2026 – Jul 2026

AI Research Tooling Intern | SHL, Gurgaon (Remote)

[GitHub](#) | [Live](#)

- Built an AI research assistant for querying academic papers, retrieving evidence, and identifying research gaps; extracted papers into schema-locked JSON and indexed 1,200-character chunks with 200 overlap using 384-dim BAAI/bge-small-en-v1.5 embeddings.
- Implemented RAG, vector search, cosine similarity, and retrieval-augmented question answering using top-6 hits with an explicit “not in this paper” refusal; added multi-paper gap analysis, hypothesis generation, and literature synthesis.

Projects

PathoQuant — Nuclei Segmentation & Quantification

[GitHub](#)

- Trained a residual U-Net (Dice + BCE) on MoNuSeg (30 H&E tiles, $\sim 21.6\text{k}$ nuclear boundaries, 7 organs) and reported Dice 0.81 / AJI 0.59 on a held-out split.
- Designed overlap-tile inference with Gaussian blending so a 256×256 model runs on $\sim 50\text{k} \times 50\text{k}$ WSIs at constant VRAM ($\sim 1.2\text{ GB}$) instead of loading the full slide.
- Quantified nuclei count, area, and density heatmaps from predicted instance masks, and beat Otsu + watershed by ~ 0.18 AJI on touching / overlapping nuclei.

HistoMIL — Gigapixel WSI Tumor Classification

[GitHub](#)

- Trained gated Attention-MIL (ResNet-style tile encoder) on Camelyon-style H&E bags to classify tumor vs. normal at slide level without pixel labels; AUC 0.91 on held-out slides.
- Tiled gigapixel WSIs into 256×256 tissue-only patches (HSV filter), aggregated 10k–40k tiles/slide, and used attention weights as a localization heatmap for tumor regions.
- Exported ONNX and ran tile inference at $\sim 12\text{ ms/tile}$ (RTX 3050) for a cloud-style, real-time pathology classification path.